Government Engineering College Kishanganj

Department of Computer Science and Engineering

Computer Networks LAB-02

Exploring Packet Sniffer and Packet Analyzer

This part of the assignment aims to understand the network packet structures and different fields present at the packet headers corresponding to different layers of the protocol stack. For this assignment, you must install Wireshark on your machine and keep your machine connected to the Internet.

Steps to Follow

I. Ensure that no browsing activity is going on in your machine. Close all browser tabs.

II. Download and run Wireshark.

III. Start capturing packets over your default (active) network interface.

IV. Open a browser tab and visit: `http://neverssl.com/`

V. Wait for 2 minutes.

VI. Now on the same browser tab, visit: `https://www.cornell.edu/`

VII. Close the browser tabs.

VIII. Stop packet capture.

IX. Save the `.pcap` file from Wireshark.

Questions

1. **[3 marks]** What are the different protocols you observe at the following layers of the protocol stack?

   A. Application layer

   B. Transport layer

   C. Network layer

2. **[2 marks]** What is the total amount of data being received for the following two cases?

   A. When you access `http://neverssl.com/`

   B. When you access `https://www.cornell.edu`

3. [**6 marks**] Answer the following when you access the site `http://neverssl.com/`.

   A. How many HTTP GET requests do you observe? List down the GET requests.

   B. What version of HTTP the server is running?

   C. For each of the HTTP GET requests, find out:

      (i) The total number of TCP segments being received

      (ii) The total amount of data being received in the corresponding HTTP response message

   D. Identify TCP three way handshake used to establish the connection with the http://neverssl.com . Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.